

Nam Hee Kim

CURRICULUM VITAE

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Education

PhD - Embodied Intelligence · Computational Behavior

Helsinki, Finland

AALTO UNIVERSITY, DEPARTMENT OF COMPUTER SCIENCE

2021 - 2026

- Supervisors: Perttu Hämäläinen · Jaakko Lehtinen · Antti Oulasvirta
- Research interests: physics-based character animation · deep reinforcement learning · generative models
- Thesis (IN PROGRESS): Learning to Motion Plan—Advances in Algorithms, Architectures, and Applications
- NSERC Canada Graduate Scholarship-Doctoral (CGS-D) Award · Finnish Center for AI Open Application Award

MSc - Motion Control · Character Animation

Vancouver, Canada

UNIVERSITY OF BRITISH COLUMBIA, DEPARTMENT OF COMPUTER SCIENCE

2019 - 2021

- Supervisor: Michiel van de Panne
- Thesis: Directable Physics-based Character Animation
- NSERC Canada Graduate Scholarship-Master's (CGS-M) Award · Graduate Teaching Assistant Award

BSc - Bioinformatics · Data Science

Vancouver, Canada

UNIVERSITY OF BRITISH COLUMBIA, DEPARTMENT OF COMPUTER SCIENCE

2016 - 2018

- Ranked #1 in the Combined Major program cohort

BSc - Human-Computer Interaction · Bioinformatics

Houston, United States

RICE UNIVERSITY, DEPARTMENT OF COMPUTER SCIENCE, TRANSFERRED

2012 - 2016

- President's Honor Roll

Publications

10. **Kim, N. H.**, Liu, J. M., Lehtinen, J., Hämäläinen, P., O'Brien, J.F., & Peng, X.B. (2026). Robo-Saber: Generating and Simulating Virtual Reality Players. Computer Graphics Forum (Eurographics 2026).
9. Cheema, N., Xu, R., **Kim, N. H.**, Hämäläinen, P., Golyanik, V., Habermann, M., Theobalt, C. & Slusallek, P. (2023). Discovering Fatigued Movements for Virtual Character Animation. In ACM SIGGRAPH Asia 2023 Conference Proceedings (pp. 1-12).
8. Mensah, D., **Kim, N. H.**, Aittala, M., Laine, S. & Lehtinen, J. (2023). A Hybrid Generator Architecture for Controllable Face Synthesis. In ACM SIGGRAPH 2023 Conference Proceedings (pp. 1-10).
7. Cohan, S., **Kim, N.H.**, Rolnick, D. & van de Panne, M. (2022). Understanding the Evolution of Linear Regions in Deep Reinforcement Learning. In Proceedings of Neural Information Processing Systems (NeurIPS) 2022.
6. **Kim, N.H.**, Kirjonen, M. & Hämäläinen, P. (2022). Learning High-Risk High-Precision Motion Control. ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG) 2022.
5. Ng, B., Casazza, W., **Kim, N.H.**, Wang, C., Farhadi, F., Tasaki, S., Bennett, D., De Jager, P., Gaiteri, C. & Mostafavi, S.. (2021). Cascading Epigenomic Analysis for Identifying Disease Genes from the Regulatory Landscape of GWAS Variants. PLOS Genetics.
4. **Kim, N.H.**, Ling, H.Y., Xie, Z. & van de Panne, M.. (2021). Flexible Motion Optimization with Modulated Assistive Forces. In Proc. ACM Transactions on Graphics and Interactive Techniques (SCA 2021).

3. Xie, Z, Ling, H.Y., **Kim, N.H.**, & van de Panne, M.. (2020). ALLSTEPS: Curriculum-driven Learning of Stepping Stone Skills. In Proc. ACM SIGGRAPH / Eurographics Symposium on Computer Animation. **Best Paper Award**
2. **Kim, N.H.**, Xie, Z. & van de Panne, M.. (2020). Learning to Correspond Dynamical Systems. Proceedings of the 2nd Conference on Learning for Dynamics and Control (L4DC), in Proc. Machine Learning Research (PMLR) 120:105-117. **Oral Presentation (Top 14 Papers of 131 Submissions)**
1. Ng, B., **Kim, N.H.**, Thirion, B., Poline, J., Greicius, M., Mostafavi, S., & IMAGEN Consortium. (2018). Associating IQ to Brain Networks with Manifold Regression. 2018 Organization for Human Brain Mapping Annual Meeting. (Peer-Reviewed by Abstract)

Selected Media Coverage

Why Teach An AI To Climb Stepping Stones? - Two Minute Papers	2021
https://www.youtube.com/watch?v=ZZ-k0Rb8grA	
Every Step They Took led to SCA Best Paper Award - University of British Columbia	2021
https://www.cs.ubc.ca/news/2021/01/every-step-they-took-led-sca-best-paper-award	
Researchers Develop Hard-to-Hack Voting Machine - Voice of America	2016
https://www.voanews.com/usa/us-politics/researchers-develop-hard-hack-voting-machine	

Honors & Awards

NSERC Canada Graduate Scholarship-Doctoral Program	2022
CAD 35,000/yr award to promote continued excellence in Canadian research; converted to PGS-D for study-abroad (CAD 21,000/yr)	
Finnish Center for AI (FCAI) Open Application Program-Doctoral Funding	2021
Full 4-year funding provided to exceptional doctoral students interested in tackling challenges in machine learning and in artificial intelligence	
UBC Department of Computer Science Graduate Teaching Assistant Award	2021
Awarded to graduate teaching assistants recommended by faculty and students for excellent teaching efforts.	
NSERC Canada Graduate Scholarship-Master's Program	2020
CAD 17,500 award for students who demonstrate a high standard of achievement in undergraduate and early graduate studies	
UBC Department of Computer Science Academic Award of Excellence (Combined Majors)	2019
Awarded to the individual graduating with the highest academic average in combined major program	
NSERC Undergraduate Student Research Award	2017
Canadian federal fellowships to support academic research as an undergraduate student	
UBC Science Scholar / Dean's Honour List	2017, 2019
Awarded to full-time students with winter session GPA above 90%	
Rice University Undergraduate Research Assistantship	2016
Rice University's research award to support students pursuing research projects outside their core curriculum	
Rice University President's Honor Roll	2015
Awarded to students in top 25% GPA percentile (above 3.882)	

Teaching Experience

Lecturer: CS-E400211 “Research Seminar in Embodied Intelligence” - Aalto University 2024

Research seminar for graduate students on intelligent systems with movement capabilities. Master’s-level independent literature review and research course.

Tutor: CS-E4000/E4875 “Research Seminar in Computer Science” - Aalto University 2022-2024

Student supervision via weekly project meetings and personalized lectures. Master’s-level independent literature review and research course. 6 students conferred. Sessions taught: 2022, 2023, 2024.

Co-Instructor: DOM-E5129 “Intelligent Computational Media” - Aalto University 2022,2024

Lectures, assignments, student supervision. Graduate-level applied AI course for art and games. Sessions taught: 2022T3, 2024T3.

Lecturer: CPSC 340 “Machine Learning and Data Mining” - UBC 2021

Sessional lectureship, teaching UBC’s main undergraduate-level machine learning course. 3rd-year level, 160 students, 10 TAs. Sessions taught: 2021S.

TA: CPSC 440/540 “Advanced Machine Learning” - UBC 2021

Teaching Assistant for the graduate level machine learning course, taught by Prof. Mark Schmidt. Sessions taught: 2020W2.

TA: CPSC 330 “Applied Machine Learning” - UBC 2020

Teaching Assistant for the pilot session of the applied machine learning course, taught by Prof. Mike Gelbart. Sessions taught: 2019W2.

TA: CPSC 340 “Machine Learning and Data Mining” - UBC 2018-2019

Teaching Assistant for UBC’s main undergraduate-level machine learning course, taught by Prof. Mark Schmidt, Prof. Frank Wood, and Prof. Mike Gelbart. Responsible for running tutorials, holding office hours, and grading assignments/exams. Sessions taught: 2018W1, 2018W2, 2019W1

Positions & Appointments Held

Research Group Leader - Embodied Intelligence *Helsinki, Finland*

AALTO UNIVERSITY

Sep. 2023 - Dec. 2024

- Founded Aalto MIME Lab - Machine Intelligence for Motion and Embodied Applications
- Lead and organize weekly meetings, provide hands-on project supervision and mentorship, recruit thesis and seminar students
- Supervised 3 MSc theses, 1 BSc thesis.

R&D Scientist Intern - Character Animation • Foundation Models *Helsinki, Finland*

UBISOFT LA FORGE / REDLYNX

Feb. 2023 - Feb. 2024

- Developed reinforcement learning algorithms to enable a rapid motion prototyping system for non-humanoid 3D characters
- Investigated general-purpose use of LoRA-adapted LLMs for goal-driven character control

Lecturer - Computer Science *Vancouver, Canada*

UNIVERSITY OF BRITISH COLUMBIA

May 2021 - Jun. 2021

- CAD 15,000 sessional lectureship for CPSC 340: Machine Learning and Data Mining for the 2021S term.
- Carried out teaching duties in full, including lectures, assignments, exams, and administration.

Applied Scientist Intern - Machine Learning

AMAZON WEB SERVICES, AMAZON.COM, INC.

Vancouver, Canada

May 2020 - Aug. 2020

- Implemented and benchmarked active learning workflow for pedestrian detection by integrating OpenMMLab into AWS SageMaker

Software Development Engineer Intern

AMAZON WEB SERVICES, AMAZON.COM, INC.

Vancouver, Canada

May 2018 - Aug. 2018

- Optimized RDS Oracle's point-in-time restore feature by leveraging dynamic size batch submission algorithm
- Achieved six-fold increase in average throughput from improved feature
- Received full-time return offer

Research Assistant - Bioinformatics

UBC CENTRE FOR MOLECULAR MEDICINE AND THERAPEUTICS

Vancouver, Canada

Sep. 2017 - Apr. 2018

- Mentored by Prof. Sara Mostafavi and Dr. Bernard Ng. Funded by NSERC Undergraduate Research Award
- Designed parametric testing methods for non-Euclidean statistical models associating genotypes with brain co-activations
- Implemented large-scale data preprocessing and imputation pipeline involving denoising, alignment, distance mapping, and compression of genetic data

Software Development Engineer Intern

DELIVERY EXPERIENCE, AMAZON.COM, INC.

Vancouver, Canada

Jun. 2017 - Aug. 2017

- Designed and implemented internal editing tool and abstraction logic for customizing and localizing Delivery Benefit Programs in accordance with company directives
- Completed required deliverables and bonus features and deployed product to Amazon's official development platform
- Received return offer for May 2018

Research Assistant - Bioinformatics

BAYLOR COLLEGE OF MEDICINE CENTER FOR GENOME ARCHITECTURE

Houston, United States

Jan. 2016 - Apr. 2016

- Mentored by Prof. Erez Lieberman Aiden
- Developed visualization components of DNA sequencing quality control pipeline to detect physical artifacts in Illumina sequencing experiments

Research Assistant - Human-Computer Interaction

RICE UNIVERSITY COMPUTER SECURITY LAB

Houston, United States

Jan. 2016 - Apr. 2016

- Mentored by Prof. Dan Wallach and Prof. Philip Kortum. Stipend-based Research Assistantship
- Designed, built, and tested the second generation model of STAR-Vote involving sheet-feeding mechanism, audio alert system, barcode detection, and graphical user interface

Frontend Developer Intern

MEDICAL INFORMATICS CORP

Houston, United States

May. 2013 - Aug. 2013

- Medical application / healthcare data management software startup
- Built JavaScript product prototype for live patient monitoring modules for product pitch in startup partnership and sponsorship meetings
- Received return offer

Additional Positions and Activities

Technical Paper Reviewer

SIGGRAPH, EUROGRAPHICS, SPRINGER NATURE, IEEE TVCG, ICCV

2022 - PRESENT

- Technical paper program reviewer in character animation and motion control

Music Director • Peer Advisor

TAPESTRY CHURCH AT MARPOLE

*Vancouver, Canada**2017 - 2020*

- Ministerial work involving coordination for weekly gathering, counseling, music practice and performance

Athlete • Competition Announcer

UBC WEIGHTLIFTING AND POWERLIFTING CLUB

*Vancouver, Canada**2016 - 2019*

- In association with British Columbia Powerlifting Association (BCPA)

Conference Staff

IEEE INTELLIGENT VEHICLES SYMPOSIUM

*Seoul, Republic of Korea**2015*

- Facilitated 5-day conference providing interpretation and technical support

Translator • Squad Leader

REPUBLIC OF KOREA ARMY

*Seoul, Republic of Korea**2013 - 2015*

- English-Korean translation and interpretation duties involving communications and computing devices for the United States Forces Korea.

Actor • Ensemble Member

RICE UNIVERSITY MUSICAL THEATRE

*Houston, United States**2012, 2013, 2015*

- Participated in the production of: The Drowsy Chaperone (2012), The Producers (2013), and a theatrical adoption of Black Mirror (2015)